

Improving Association Rule Mining using Clustering-Based Item Similarity

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March 12, 2025

1 Abstract

Association rule mining (ARM) is commonly used to discover relationships in large datasets, but traditional methods like Apriori and FPGrowth face challenges with computational cost and rule quality.

This paper presents a clustering-based ARM approach that integrates vectors similarity to reduce the search space and focus on meaningful rules. Using KMeans clustering, our method identifies relationships that traditional frequency-based methods may overlook. We evaluate our approach on four datasets and compare it to the Apriori algorithm.

Results show that our clustering-based approach improves rule quality while maintaining computational efficiency, making it a scalable approach for ARM in large datasets.

2 Problem Description

This work aims to improve the Association Rule Mining (ARM) process, specifically enhancing lift values in traditional rule generation methods like Apriori. Conventional algorithms often generate weak rules with low lift due to relying solely on item frequency and treating transactions independently, which ignores transaction similarity. Additionally, they may overlook valuable rare rules due to computational constraints.

Our approach clusters transactions based on similarity, improving rule generation by focusing on stronger associations within clusters. This method increases lift and the quality of the generated rules, while maintaining computational efficiency.

3 Solution Overview

3.1 Dimensionality Reduction

In order to enable effective clustering and avoid the curse of dimensionality, we employ a heuristic approach for dimension reduction. We represent each transaction by selecting the top $k\%$ most popular items across the entire dataset. Our empirical analysis shows that selecting k between 1% and 20% achieves a good balance of accuracy and efficiency. This method effectively captures the core behavior or overall patterns of the transaction, while significantly reducing its dimensionality.

3.2 Clustering Using Kmeans

Our approach begins by clustering the transactions using the KMeans algorithm [5], where the number of clusters k is determined through the elbow method explained in here [6]. Since the transactional data is represented as binary vectors, KMeans is particularly suitable for this task. This is because the KMeans algorithm uses Euclidean metric which is:

$$d(A, B) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2} \quad (1)$$

in our case all of the transactions are represented as binary vectors:

$$\text{For each } i, \quad A_i = \begin{cases} 1 & \text{if item } i \text{ is present} \\ 0 & \text{o.w} \end{cases}$$

Namely, we can conclude that our Euclidean distance represents the square root of the number of items that differ between the two transactions:

$$d(A, B) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2} = \sqrt{\text{Hamming Distance}(A, B)} \quad (2)$$

Thus, transactions are grouped effectively based on similarity, where transactions with common items will be grouped together.

3.3 Applying Apriori Within Each Cluster

Once the transactions are clustered, we apply the Apriori algorithm to each cluster separately, using higher support thresholds to better suit the smaller clusters datasets and avoid noisy unreliable rules. This allows for more focused and relevant rule generation within each cluster.

3.4 FPGrowth and Apriori Equivalence

We use the FPGrowth algorithm instead of Apriori to improve computational efficiency, as it avoids costly candidate generation while still identifying the same frequent itemsets and association rules. Given their equivalence in this context [7], we use the terms interchangeably .

4 Experimental Evaluation

To assess the effectiveness of our solution, we compared it against the baseline approach using three key metrics commonly used in association rule mining: **Support**, **Confidence**, and **Lift**. These metrics provide insights into the strength and relevance of the generated association rules. Our objective was to demonstrate that our proposed approach enhances rule quality compared to the baseline.

4.1 Datasets

We conducted experiments on four publicly available datasets, transformed into transactional formats for association rule mining:

- **MovieLens 20M**: 20 million ratings from 138,493 users across 27,278 movies, converted into a transactional format by treating each user’s ratings as a transaction (available at <https://www.kaggle.com/datasets/grouplens/movielens-20m-dataset>).

- **Online Retail Dataset:** Purchase records from 4,000 customers with over 25,000 transactions, transformed by aggregating each customer’s shopping history (available at <https://www.kaggle.com/datasets/vijayuv/onlineretail>).
- **Last.FM Dataset:** 92,800 artist listening records from 1,892 users, treated as transactions based on each user’s listening history (available at <https://grouplens.org/datasets/hetrec-2011/>).
- **Netflix Prize Data:** Movie-watching histories of 480,189 users, converted into transactions by grouping all movies watched by a user (available at <https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data>).

4.2 Evaluation Methods

For each dataset, we applied the following methods:

- **Baseline Approach:** The traditional Apriori algorithm, which extracts rules from the entire dataset without considering transaction similarity.
- **Proposed Approach:** Our method, which clusters transactions using KMeans and applies the Apriori algorithm separately within each cluster, using higher support thresholds tailored to the smaller clustered datasets.

4.3 Results

4.3.1 MovieLens 20M Dataset

Table 1: Baseline Approach (min support = 0.18)
Number of rules: 3176

Metric	Avg	Min	Max
Support	0.200921	0.18	0.3542
Confidence	0.625521	0.3605	0.9566
Lift	1.8707	1.1664	3.7134

Table 2: Our Solution (in-cluster min support = 0.5)
Number of rules: 9484

Metric	Avg	Min	Max
Support	0.1965	0.0354	0.34865
Confidence	0.6443	0.1025	0.981992
Lift	2.3092	1.2140	7.6053

Our approach increases the Lift from 1.87 to 2.31, indicating stronger relationships between items.

4.3.2 Online Retail Dataset

Table 3: Baseline Approach (min support = 0.05)
Number of rules: 240

Metric	Avg	Min	Max
Support	0.0580349	0.05	0.0762791
Confidence	0.570955	0.252619	0.965957
Lift	5.67581	1.48258	11.8352

Table 4: Our Solution (in-cluster min support = 0.4)
Number of rules: 1852

Metric	Avg	Min	Max
Support	0.0439065	0.0253488	0.0755814
Confidence	0.572033	0.129505	1.0000000
Lift	8.22112	2.12599	29.2759

Our approach shows a noticeable increase in Lift (from 5.68 to 8.22).

4.3.3 Last.FM Dataset

Our approach results in an increase in Lift (from 3.61 to 5.50) and a decrease in Confidence. The lower Confidence suggests that some rules are less certain.

Table 5: Baseline Approach (min support = 0.099)

Number of rules: 4458

Metric	Avg	Min	Max
Support	0.1159801	0.1002191	0.2387733
Confidence	0.6656595	0.2995090	0.9947368
Lift	3.6084335	1.6627986	5.0973262

Table 6: Our Solution (in-cluster min support = 0.4)

Number of rules: 2242

Metric	Avg	Min	Max
Support	0.0238546	0.0093100	0.2168675
Confidence	0.3884072	0.0278232	1.0000000
Lift	5.5031059	0.6973268	17.1993721

4.3.4 Netflix Prize Dataset

Table 7: Baseline Approach (min support = 0.09)

Number of rules: 112

Metric	Avg	Min	Max
Support	0.0330308	0.0108033	0.3456590
Confidence	0.7132389	0.0434620	1.0000000
Lift	21.4071420	2.8930248	92.5643777

Table 8: Our Solution (in-cluster min support = 0.02)

Number of rules: 616

Metric	Avg	Min	Max
Support	0.0060682	0.0010896	0.3456590
Confidence	0.3343641	0.0031522	1.0000000
Lift	37.0355164	0.1391973	917.7659574

Our approach results in a dramatic increase in Lift (from 21.41 to 37.04). In addition, as the results above, we observe again a decrease in Confidence.

4.3.5 Summary

Our approach outperforms the baseline by generating association rules with higher lift, while taking no more time than the baseline. In addition it is at least as scalable as the Apriori algorithm, producing more rules in 3 out of 4 datasets. Although our results tend to have lower overall support, the minimum support threshold within each cluster ensures that all generated rules have strong support within their similarity group.

5 Related Work

Previous studies, like Dharshinni [1], applied KMeans clustering based on attributes such as gender and age, improving rule quality when combined with the Apriori algorithm. M. Harahap et al. [4] used clustering to identify patterns in disease data. However, these studies mainly compared rules within clusters. In contrast, our approach evaluates and compares the rule sets generated by the baseline and our method as a whole, demonstrating the superior quality of our approach.

Inspired by user-based collaborative filtering (CF) methods [2], we cluster users based on transaction history, avoiding reliance on demographic information and offering a cluster-based solution when no prior knowledge is present.

6 Conclusion

In this paper, we presented a novel approach to improving Association Rule Mining (ARM) by incorporating clustering-based item similarity. By clustering transactions based on their similarity, we effectively reduced the search space, allowing the Apriori algorithm to generate more meaningful and relevant association rules. Our experiments demonstrated that this ap-

proach significantly enhances the quality of generated rules, particularly in terms of Lift, indicating stronger relationships between items.

The results from multiple datasets showed that our method not only improved rule quality but also maintained computational efficiency, making it a scalable solution for large datasets. Overall, our clustering-based approach offers a powerful enhancement to traditional ARM methods, providing an effective alternative to discover valuable item relationships in complex datasets.

References

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